

Lecture 09: Instrumental variables II

PPHA 34600
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From last time: introduction to IV

Recall that we want to split $D_i = B_i\varepsilon_i + C_i$ into the C_i and other parts

An instrumental variable...:

...Generates variation in C_i but is uncorrelated with ε_i

Z_i is a valid instrument for D_i when the following are satisfied:

① **First stage:** $\text{Cov}(Z_i, D_i) \neq 0$

- Z_i and D_i are related
- Without this, you're capturing nothing
- This is actually testable!

② **Exclusion restriction:** $\text{Cov}(Z_i, \varepsilon_i) = 0$

- Z_i and ε_i are **not** related
- Z_i only affects Y_i through D_i
- Fundamentally untestable!

What makes IV so useful?

IV can be used in many ways:

- Causal inference (see last time)
- (Omitted variable bias)
- Measurement error

An omitted variable bias refresher

Suppose the true data generating process is:

$$Y_i = \alpha + \tau D_i + \beta X_i + \varepsilon_i$$

We'll assume:

- D_i and X_i are uncorrelated with ε_i
- D_i and X_i are correlated with each other
→ $\text{Cov}(D_i, X_i) \neq 0$
- We don't observe X_i ()
→ Now we have to run:

$$Y_i = \alpha + \tau D_i + \nu_i$$

where

$$\nu_i = \varepsilon_i + \beta X_i$$

An omitted variable bias refresher

$$\begin{aligned}\hat{\tau} &= \frac{\text{Cov}(Y_i, D_i)}{\text{Var}(D_i)} \\ &= \underbrace{\frac{\text{Cov}(\alpha + \tau D_i + \beta X_i + \varepsilon_i, D_i)}{\text{Var}(D_i)}}_{\text{plug in for } Y_i}\end{aligned}$$

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With omitted variable bias, we instead have

$$\hat{\tau} = \tau + \beta \frac{\text{Cov}(D_i, X_i)}{\text{Var}(D_i)} \neq \tau$$

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- Suppose we have an instrument, Z_i
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- In this case, the error term is $\nu_i = \varepsilon_i + \beta X_i$
 - In other words, $\text{Cov}(Z_i, D_i) \neq 0$ and $\text{Cov}(Z_i, \nu_i) = 0$

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- The second-stage IV estimate is then:

$$\hat{\tau}^{2SLS} = \tau + \beta \frac{\text{Cov}(\hat{D}_i, X_i)}{\text{Var}(\hat{D}_i)}$$

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$$\hat{\tau}^{2SLS} = \tau + \underbrace{0}_{\text{exclusion restriction}}$$

We often worry about measurement error:

- What happens if we don't perfectly observe D_i or Y_i ?
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- The answer is...
- It depends!

Consider a true relationship

True relationship: $\tau = -0.059$

Measurement error in Y is fine

Estimated relationship: $\hat{\tau} = -0.061$

Why does this work?

We don't observe Y_i , but rather $\tilde{Y}_i = Y_i + \gamma_i$

→ Assume $\text{Cov}(\gamma_i, \varepsilon_i) = 0$ and $\text{Cov}(\gamma_i, D_i) = 0$

If we run:

$$\tilde{Y}_i = \alpha + \tau D_i + \varepsilon_i$$

We'll estimate:

$$\hat{\tau} = \frac{\text{Cov}(\tilde{Y}_i, D_i)}{\underbrace{\text{Var}(D_i)}_{\text{def'n of OLS}}}$$

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Success!

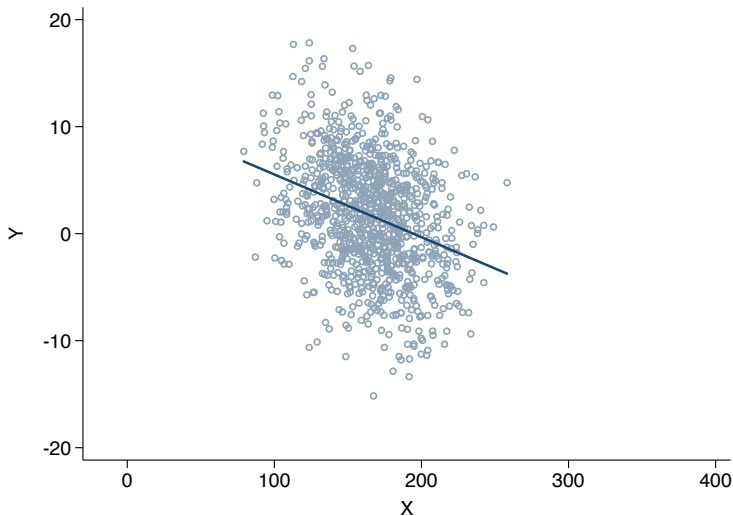
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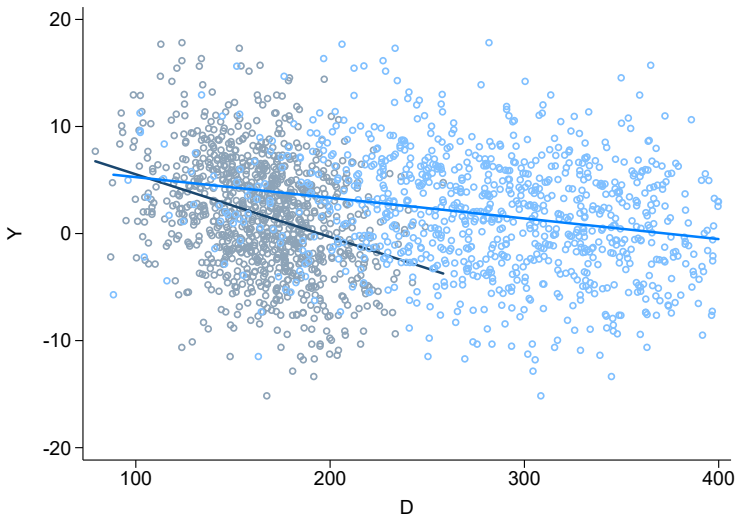
Estimated relationship: $\hat{\tau} = -0.015$

Non-classical measurement error in D_i is bad



True relationship: $\tau = -0.059$

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Estimated relationship: $\hat{\tau} = -0.019$

What's going wrong?

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What's going wrong?

$$\hat{\tau} = \tau \left(\frac{\text{Var}(D_i)}{\text{Var}(D_i) + \text{Var}(\gamma_i)} \right)$$

This is the classic **attenuation bias**

- $\hat{\tau}$ is biased towards zero
- Note we assumed the most innocuous form of measurement error
- If measurement error is correlated with treatment, we get OVB

A second trip to the instrument store

To solve the measurement error problem, we'll use a clever instrument:

- We will instrument for $\tilde{D}_i$ with Z_i , a different noisy measure of D_i :

$$Z_i \equiv \tilde{D}_i = D_i + \zeta_i$$

Assume:

- $Cov(\zeta_i, D_i) = 0$: Measurement error is uncorrelated with treatment
- $Cov(\zeta_i, \gamma_i) = 0$: Measurement error in Z_i is uncorrelated w error in $\tilde{D}_i$
- $Cov(\zeta_i, \varepsilon_i) = 0$: Measurement error is uncorrelated with original error

Does this meet our two assumptions?

- ➊ **First stage:** Yes! $Cov(Z_i, \tilde{D}_i) \neq 0$
- ➋ **Exclusion restriction:** Yes! $Cov(Z_i, \varepsilon_i) = 0$

What do these assumptions buy us?

Remember that:

$$\hat{\tau}^{IV} = \frac{\text{Cov}(Y_i, Z_i)}{\text{Cov}(\tilde{D}_i, Z_i)}$$

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Success!

IV solves measurement error

What's the intuition?

- $\tilde{D}_i = D_i + \gamma_i$
 - $Z_i = \dot{D}_i = D_i + \zeta_i$
- Z_i and $\tilde{D}_i$ only have the **true** D_i in common
- We've assumed that $\text{Cov}(\gamma_i, \zeta_i) = 0$

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The first stage is:

$$\tilde{D}_i = \alpha + \pi \mathring{D}_i + \epsilon_i$$

- We're only using the variation from D_i (not from ζ_i or γ_i)!

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- We're only using the variation from D_i (not from ζ_i or γ_i)!
- **Important caveat:** This does not work with binary D_i !
- If true $D_i = 1$, measurement error can only be -1 or 0
 - If true $D_i = 0$, measurement error can only be 0 or 1
- Measurement error in $\tilde{D}_i$ and $\mathring{D}_i$ will be correlated

Non-classical measurement error

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Assume:

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Relax the orthogonality assumption:

- Allow $\text{Cov}(D_i, \gamma_i) \neq 0$: Measurement error can be correlated with treatment

IV with non-classical measurement error

Again, we'll now have:

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$$\hat{\tau} = \frac{\text{Cov}(\alpha, D_i) + \text{Cov}(\alpha, \gamma_i) + \tau \text{Cov}(D_i, D_i)}{\text{Var}(D_i) + \text{Var}(\gamma_i) + 2\text{Cov}(D_i, \gamma_i)} \\ + \frac{\tau \text{Cov}(D_i, \gamma_i) + \text{Cov}(D_i, \varepsilon_i) + \text{Cov}(\gamma_i, \varepsilon_i)}{\text{Var}(D_i) + \text{Var}(\gamma_i) + 2\text{Cov}(D_i, \gamma_i)}$$

IV with non-classical measurement error

$$\begin{aligned}\hat{\tau} &= \frac{\text{Cov}(\alpha, D_i) + \text{Cov}(\alpha, \gamma_i) + \tau \text{Cov}(D_i, D_i)}{\text{Var}(D_i) + \text{Var}(\gamma_i) + 2\text{Cov}(D_i, \gamma_i)} \\ &+ \frac{\tau \text{Cov}(D_i, \gamma_i) + \text{Cov}(D_i, \varepsilon_i) + \text{Cov}(\gamma_i, \varepsilon_i)}{\text{Var}(D_i) + \text{Var}(\gamma_i) + 2\text{Cov}(D_i, \gamma_i)} \\ &= \tau \underbrace{\left(\frac{\text{Var}(D_i) + \text{Cov}(D_i, \gamma_i)}{\text{Var}(D_i) + \text{Var}(\gamma_i) + 2\text{Cov}(D_i, \gamma_i)} \right)}_{\text{rearrange}}\end{aligned}$$

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- Again, we get **bias**
- Note that this **need not attenuate** $\hat{\tau}$
- This can actually **flip the sign** of $\hat{\tau}$ relative to τ
- (This depends on the sign of $\text{Cov}(D_i, \gamma_i)$)

What about IV?

Just like before...:

$$\begin{aligned}\hat{\tau}^{IV} &= \frac{\text{Cov}(Y_i, Z_i)}{\text{Cov}(\tilde{D}_i, Z_i)} \\&= \underbrace{\frac{\text{Cov}(\tau D_i + \varepsilon_i, Z_i)}{\text{Cov}(D_i + \gamma_i, Z_i)}}_{\text{definition of } Y_i, \tilde{D}_i} \\&= \underbrace{\frac{\tau \text{Cov}(D_i, Z_i) + \text{Cov}(\varepsilon_i, Z_i)}{\text{Cov}(D_i, Z_i) + \text{Cov}(\gamma_i, Z_i)}}_{\text{variance rules}} \\&= \underbrace{\tau \left(\frac{\text{Cov}(D_i, Z_i)}{\text{Cov}(D_i, Z_i)} \right)}_{\text{assumptions}} \\&= \tau\end{aligned}$$

Success!

An example: Early-life rainfall and health

Policy issue:

- Early-life shocks may be very important
- With bad harvests, kids may not get the proper nutrition

Approach:

- (We're not actually evaluating a program here)
- We want to estimate the effect of rainfall on health
- **Measurement of rainfall is poor in Indonesia**
- Instrument of choice: rainfall at weather stations $j \neq i$

Estimating the effects of rainfall on health

The authors will run a (simplified) version of:

$$Y_i = \tau \text{Rainfall}_i + \varepsilon_i$$

Where:

Y_i is a health outcome of interest

Rainfall_i is rain in location i

- (They'll actually do this in a series of lags)

ε_i is an error term

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A big concern

- Rainfall_i is measured with error
- We are likely to understate the true effect
- **Solution:** $Z_i = \text{Rainfall Nearby}_i!$

First stage estimates

Dependent variable: Rainfall in birthyear and birthdistrict (deviation of log rainfall in birth district from log of 1953-1999 district mean rainfall)

	<u>Women</u>	<u>Men</u>
Birthyear/birthdistrict rainfall, 2nd-closest station	0.138 (0.024)***	0.120 (0.023)***
Birthyear/birthdistrict rainfall, 3rd-closest station	0.144 (0.039)***	0.158 (0.035)***
Birthyear/birthdistrict rainfall, 4th-closest station	0.088 (0.053)	0.081 (0.044)*
Birthyear/birthdistrict rainfall, 5th-closest station	0.125 (0.025)***	0.158 (0.039)***
Number of observations	4,615	4,277
R-squared	0.59	0.59
F-statistic: Joint significance of all four rainfall variables	31.61	28.80
P-value	0.000	0.000

Placebo test estimates

Coefficients (std. errors) in regression of outcome on *child's* birthyear rainfall.

	<u>Women</u>	<u>Men</u>
<u>Mother's characteristics</u>		
Completed grades of schooling	0.204 (1.136) [2,447]	0.132 (0.947) [2,258]
Currently alive (indicator)	0.084 (0.083) [4,542]	0.029 (0.108) [4,039]
<u>Father's characteristics</u>		
Completed grades of schooling	0.273 (1.172) [2,810]	0.166 (1.309) [2,621]
Currently alive (indicator)	0.010 (0.080) [4,541]	-0.093 (0.169) [4,040]

Placebo test estimates

	<u>Women</u>	<u>Men</u>
Self-rep. health status very good (indic.)	0.123 (0.099) [1,239]	-0.115 (0.078) [1,264]
Self-rep. health status poor/very poor (indic.)	0.090 (0.154) [1,239]	0.106 (0.134) [1,264]
Ln (lung capacity)	-0.067 (0.034)* [1,195]	0.008 (0.089) [1,130]
Height (cm.)	-1.165 (1.660) [1,207]	3.054 (2.017) [1,132]
Days absent due to illness (last 4 weeks)	0.669 (0.688) [1,240]	3.075 (1.505)* [1,261]
Completed grades of schooling	0.958 (1.274) [1,240]	-1.441 (1.947) [1,260]
Ln (expenditures per cap. in hh)	-0.193 (0.284) [1,240]	-0.329 (0.189) [1,264]
Asset index	-0.773 (0.497) [1,240]	0.166 (0.353) [1,264]
Ln (annual earnings)	0.202 (0.333) [631]	-0.612 (0.344) [1,142]

2SLS estimates

TABLE 2—EFFECT OF BIRTH YEAR RAINFALL ON ADULT OUTCOMES: WOMEN AND MEN BORN 1953–1974
(Instrumental variables estimates. Coefficients (standard errors) in regression of outcome on rainfall in individual's birth year and birth district. Instrumental variables for birth year/birth district rainfall are rainfall measured at second-through fifth-closest rainfall stations to respondent's birth district.)

	Women	Men
Self-reported health status very good (indicator)	0.101 (0.058)* [4,613]	−0.029 (0.072) [4,270]
Self-reported health status poor/very poor (indicator)	−0.192 (0.082)** [4,613]	−0.100 (0.098) [4,270]
Ln (lung capacity)	−0.044 (0.049) [4,454]	−0.073 (0.062) [3,907]
Height (centimeters)	2.832 (0.821)*** [4,495]	0.998 (1.795) [3,924]
Days absent due to illness (last four weeks)	−1.175 (0.831) [4,611]	0.515 (0.779) [4,267]
Completed grades of schooling	1.086 (0.453)** [4,598]	−0.474 (1.490) [4,259]
Ln (expenditures per capita in household)	0.095 (0.204) [4,615]	−0.274 (0.301) [4,277]
Asset index	0.876 (0.324)** [4,613]	−0.279 (0.507) [4,276]
Ln (annual earnings)	0.065 (0.988) [2,332]	−0.202 (0.350) [3,963]

2SLS estimates

TABLE 3—EFFECT OF RAINFALL IN YEARS BEFORE AND AFTER BIRTH: WOMEN BORN 1953–1974
(Instrumental variables estimates. Rainfall in individual's birth year and birth district instrumented with rainfall measured at second- through fifth-closest rainfall stations to respondent's birth district.)

Dependent variable	Self-reported health status very good (indicator)	Self-reported health status poor/very poor (indicator)	Height (centimeters)	Completed grades of schooling	Asset index
Coefficient on rainfall in:					
Year -3	0.025 (0.084)	-0.114 (0.120)	1.505 (1.572)	-0.065 (0.992)	0.003 (0.424)
Year -2	-0.037 (0.103)	-0.013 (0.075)	0.854 (1.813)	-0.852 (1.670)	-0.426 (0.721)
Year -1	-0.080 (0.123)	-0.045 (0.088)	3.338 (2.155)	0.104 (1.332)	-0.380 (0.530)
Year 0	0.090 (0.067)	-0.179 (0.093)*	3.833 (1.420)**	1.598 (0.675)**	0.750 (0.399)*
Year 1	-0.008 (0.053)	-0.096 (0.067)	0.676 (1.592)	1.083 (0.769)	0.203 (0.272)
Year 2	-0.041 (0.043)	-0.015 (0.068)	1.666 (0.984)	0.117 (0.840)	-0.229 (0.452)
Year 3	-0.020 (0.116)	-0.104 (0.067)	1.996 (1.774)	-0.135 (0.802)	0.088 (0.232)
Observations	4,613	4,613	4,495	4,598	4,613

TL;DR:

- ① Instrumental variables are very powerful
- ② With the right assumptions...
- ③ ...we can handle OVB and ME (and simultaneity)

For next class

Topics:

- Instrumental variables III

Reading: Day off!

- Take another look at Fowlie, Wolfram, et al.